

Clinical presentation of common parasitic diseases

Learning Objectives

- ⇒ To discuss the meaning of parasites
- ⇒ To classify the parasites based on their morphological structures and habitats
- ⇒ To describe pathogenesis of various parasites
- ⇒ To perform the relevant clinical examination of the patients of hookworm, round worm, thread worm and pin worm
- ⇒ To order and interpret the relevant laboratory investigations for the diagnosis of hookworm, roundworm, threadworm and pinworm

Introduction

Parasites are probably the most complex of all microbes infecting humankind. They can be unicellular organisms (e.g. *Plasmodium species*, which causes malaria) or multicellular, with sizes extending upto 10 meters (e.g. tapeworm). Their life cycles may be simple or complex, requiring multiple hosts and vectors. The literal meaning of parasite is 'eating from another's plate'. In other words, parasites take their nutrition from their hosts where they are residing.

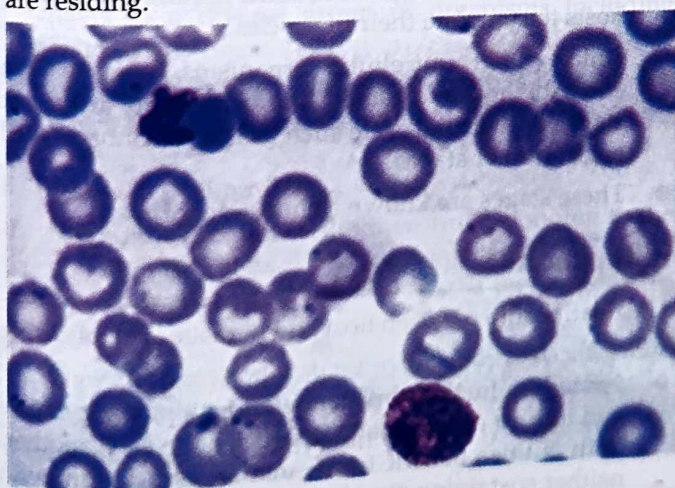


Figure 229: Species of plasmodium

Characters of parasites

The following are the characters of parasites

- ⊗ A living organism which receives nutrition and shelter from another organism where it lives is called parasite.
- ⊗ A parasite does not necessarily cause disease.
- ⊗ Simply parasitism is living in association with the host.
- ⊗ The parasite derives all benefits from the association and the host may either not be harmed or may suffer the consequences of this association, a parasite disease.
- ⊗ A parasite is an organism that is entirely dependent on another organism, referred to as its host, for all or part of the life cycle and metabolic requirements.
- ⊗ **Host:** an organism which harbours the parasite.

Classes of parasites

Parasites are mainly classified into following two categories; along with some other types of parasites:

- ⊗ **Unicellular** (having a single cell, can be visualised under microscope, e.g. protozoa such as *Entamoeba histolytica* (see Figure 230) and multicellular (can be visualised with naked eye, e.g. helminths such as roundworms (see Figure 231)
- ⊗ **Ecto-parasite (ectozoa):** Lives outside on the surface of the body of the host.
- ⊗ **Endo- parasite (entozoan):** Lives inside the body the host; in the blood, tissues, body cavities, digestive tract and other organs.

- ⊗ **Temporary parasite** – visits its host for a short period
- ⊗ **Permanent parasite** – leads a parasitic life throughout the whole period of its life.
- ⊗ **Facultative parasite** – lives a parasitic life when opportunity arises.
- ⊗ **Obligatory parasite** – cannot exist without the parasitic life.
- ⊗ **Wandering or aberrant parasite** – happens to reach a place where it cannot live.
- ⊗ **Occasional or accidental parasite** – attacks an unusual host.

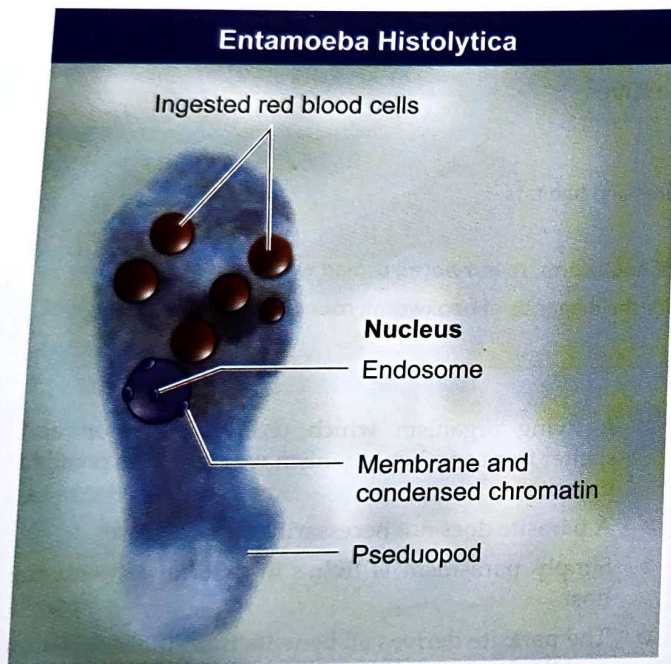


Figure 230: Entamoeba histolytica

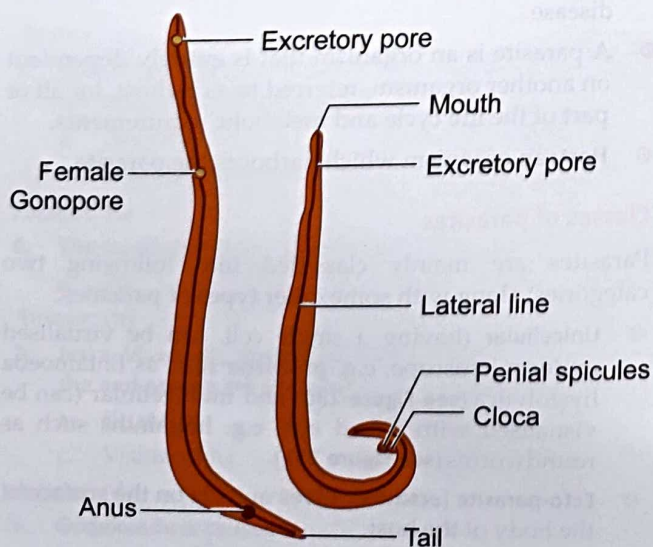


Figure 231: Roundworms

Host

A host is an entity that provides nourishment and harbours the parasite. In the process, the host may suffer injury and lose nutrients. Depending on their relationship with parasite, hosts can be categorised as follows:

Definitive host

This is a host in which adult stage of the parasite lives (e.g. helminths) or the organism in which the parasite undergoes sexual reproduction (e.g. protozoa)

Intermediate host

A host that harbours the larval stage of the parasite (helminths) or in which asexual reproduction takes place (protozoa) is known as an intermediate host. For example, humans are intermediate hosts in malaria and hydatid disease.

Reservoir host

This type of host harbours the parasite and also acts as a source of infection.

Paratenic host

The parasite is unable to develop further in this host. It can act as a transport or carrier host.

Habitat

A place where the parasite naturally lives and grows is known as its habitat, e.g. malarial parasite in humans resides in the liver and RBC.

Life cycle of parasites

- ⊗ While a number of parasites follow a simple or direct life cycle with a single host, some require two or more hosts to complete their life cycles.
- ⊗ Unlike other unicellular organisms, parasites can enter the host and establish infection only at specific stages in their life cycle such as larvae, embryonated eggs or other stages.
- ⊗ These stages are known as infective forms.

Host parasite

The relationship between host and parasite is of following types.

- ⊗ **Symbiosis:** In this type of relationship, the parasite and its host are mutually dependent on each other. Both entities benefit from such an association, with neither sustaining any damage or injury.
- ⊗ **Commensalism:** This refers to an association from which the parasite benefits, while the host is neither benefited nor harmed.

- ❖ **Parasitism:** This is a situation in which one living organism is fully dependent on the host for survival and cannot live independently.

Parasite derive the benefits of nutrition and shelter from the host they live in. In the process, the host is harmed either by loss of nutrients or by pathological changes caused by the parasite harboured in the host.

Classification of parasites

Classification of parasites which can cause disease can be classified in three kingdoms –

1. Protozoa;
2. Stramenopila and
3. Animalia.

4. Protozoa

- ❖ Protozoa are unicellular, eukaryotic, simple, microscopic parasites (e.g. Plasmodium) which obtain nutrition by ingesting nutrients present in the host.
- ❖ Intestinal protozoa are unicellular organisms with very little structural differentiation. Each contains at least one or several nuclei.
- ❖ They may be motile – on account of the presence of pseudopodia, cilia or flagella or non motile.
- ❖ The single protozoan cell performs all life functions like respiration, digestion, reproduction etc.
- ❖ The generation time is short and the reproduction rate is high. Most protozoa are harmless and exist in nature of soil and water.

5. Stramenopila

- ❖ These are unicellular, plant like animals. This term refers to a diverse group of organisms including diatoms and algae.

The pathogenic parasite in this group is *Blastocystis*, which causes gastrointestinal infections.

6. Animalia (Helminths)

- ❖ Helminths are multicellular worms belonging to the kingdom Animalia. They are elongated, flat or round worms that can range in size from a few millimetres to meters.

They affect several systems in the body and cause mild asymptomatic to severe life threatening infections.

- ❖ Helminths are classified into two phyla which are as follows:

- **Phylum Nematelminths** – includes a single group of pathogen, the nematodes.
- **Phylum Platyhelminths** – includes two classes of pathogens, namely cystodes and trematodes.

Nematodes

Characters of nematodes

- ❖ Adult worms range in size from 3 mm to 1 m.
- ❖ They are long, cylindrical, bilaterally symmetrical and unsegmented, with both ends pointed and without any appendages.
- ❖ A pseudocoel (body cavity) is present between the body wall and intestine.
- ❖ The sexes are separate and females are generally longer than males.
- ❖ The body wall has an outer tough cuticle layer and a longitudinal muscle, which is responsible for the movement of the worm.
- ❖ The alimentary canal is well-developed, with a mouth at the anterior end and a substernal anus. Hookworms have teeth or cutting plates. Internally, nematodes have a well-developed glandular esophagus, intestine, rectum.
- ❖ The reproductive system in males comprises a long, convoluted tube, which can be differentiated into the testes, vas deferens, seminal vesicle and ejaculatory duct. The ejaculatory duct opens at the posterior end along with the rectum.
- ❖ The female reproductive system consists of an ovary, oviduct, seminal receptacle and uterus; both tubes open into a vagina, which opens through a genital pore near the mouth or in the middle of the body.
- ❖ The nervous system is rudimentary, consisting of a brain which is a circular ring with six nerve trunks running longitudinally.
- ❖ The excretory system too is rudimentary, sometimes ending in an anus other times with waste being excreted through the body wall. In some members, the excretory canal is located near the oesophagus.
- ❖ Intestinal nematodes complete their life cycle in a single host.



Figure 232: Wuchereria

- ❖ Nematodes either lay eggs (oviparous e.g. Ascaris)

or larvae (viviparous e.g. *Wuchereria* see Figure 232) or can lay eggs that immediately hatch into larvae (ovoviviparous e.g. *Strongyloides*).

Clinical implications of nematode infections

- Most patients infected with intestinal helminths remain asymptomatic.
- The clinical presentation may be limited to vague abdominal pain and occasional diarrhoea.
- In children, the habit of eating of mud (pica), growth retardation and malnutrition may be the only complaints.
- The bolus of adult worms can block the intestine, causing pain abdomen, intestinal obstruction, perforation, intussusception or volvulus.
- The migration of adult worms into the biliary tract can cause biliary colic, cholecystitis, cholangitis and pancreatitis.
- Allergic manifestations such as fever, urticaria, angioneurotic oedema and conjunctivitis can occur in all nematode infections.
- Patients infected with hookworms may present with anemia and malnutrition.
- The migration of larvae through the skin may cause cutaneous manifestations.

Cestodes

Characters of cestodes

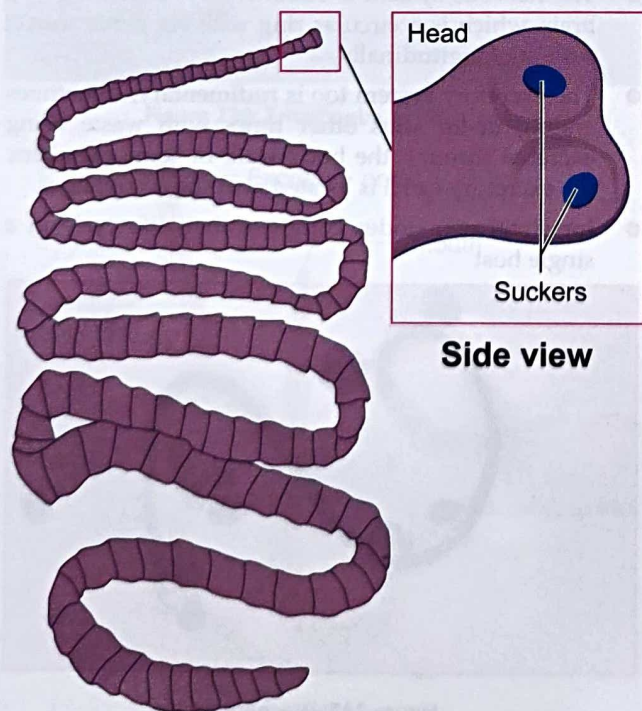


Figure 233: Tapeworms

- The adult worm, which usually resides in the intestine of its host is segmented, long and flattened dorsoventrally. Due to their tape like form, cestodes are called tape worms. (see Figure 233)
- Their body is covered with a tough cuticle layer, there is no body cavity.
- Broadly, the morphological features of the adult worm – a head (scolex), a neck and body (strobila).
- The scolex or head is the organ of attachment. It can bear suckers (as in taenia) or a pair of longitudinal grooves, called bothria. The head continues into a neck.
- The strobila or body is made up of multiple segments called proglottids. Small tapeworms like *Echinococcus granulosus* possess 3-4 segments, while longer ones like *Taenia saginata* may have upto 1,000 segments.
- These worms lack an alimentary canal; they absorb the nutrients through their outer coat.
- Excretory tubules are present in each segment and waste products are eliminated through the terminal flame cells.
- The sexes are not separate. Both male and female reproductive systems are present in each proglottid, i.e. these worms are hermaphrodites.
- The reproductive system is highly developed in cestodes. Both male and female genital openings are located in common genital pore. Fertilisation occurs between segments of same worm. Gravid proglottids containing hundreds of eggs are released into the environment by the host.
- The nervous system is primitive and consists of the cephalic ganglion in the scolex, which is connected with sensory and motor nerves across proglottids.
- In *Diphyllobothrium*, the eggs are operculated (bearing a lid), while in taenia, the eggs may contain a hexacanth embryo.

Trematodes

Characters of trematodes

- Adults are leaf like, flat, unsegmented worms ranging in size from 1 mm to several cms. Females are longer and thinner than males.
- Trematodes reside in the intestine, liver, lungs or blood vessels.
- Their body is covered with an integument: they lack a body cavity.
- Eggs are oval to round, operculated and embryonated when laid.
- Several larval forms are seen – metacercaria which are the infective forms. Cercaria, redia, sporocysts and miracidia.

- The sexes are not separate except in schistosomes.
- The alimentary tract consists of a blind, bifurcated intestinal tract without an anus.
- Trematodes have a primitive nervous system and well developed excretory and reproductive systems.
- Trematodes are oviparous. Eggs are operculated (lidded), containing a small ovum with yolk granules. Schistosoma eggs are not operculated and have a spine.
- Trematodes are present in many animals beside humans and are often zoonoses.

Clinical implications of trematodes

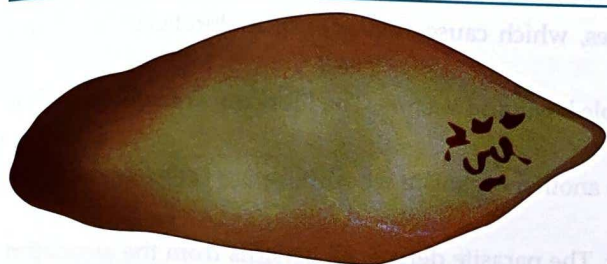


Figure 234: Liver flukes

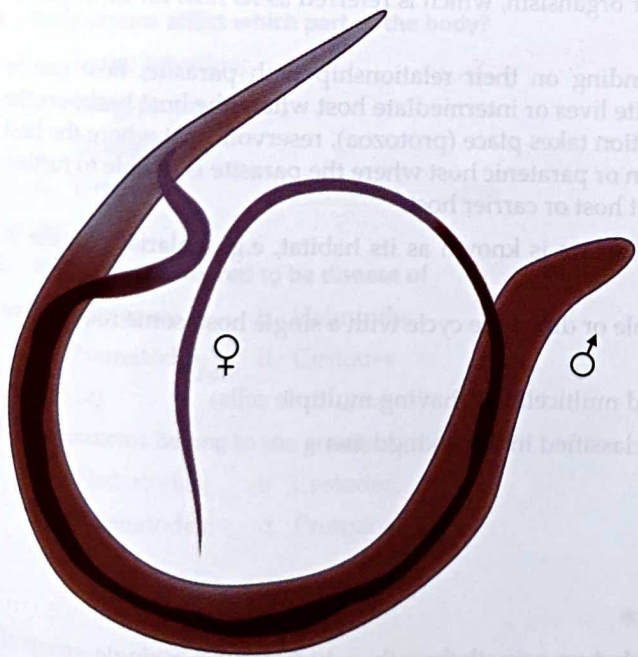


Figure 235: Blood flukes

- Trematodes comprise liver flukes (see Figure 234), blood flukes (see Figure 235), lung flukes (see Figure 236) and intestinal flukes (see Figure 237).

In the acute stages, liver flukes cause vague abdominal symptoms like abdominal pain, anorexia, nausea, diarrhoea and flatulence.

- Later, the worm causes inflammation and granuloma and abscesses, followed by fibrosis and bile duct hyperplasia, causing cholangitis, cholecystitis,

cholelithiasis, liver rot, cirrhosis and pancreatitis.

- Blood flukes can cause urogenital disease and predispose to urinary bladder carcinoma.

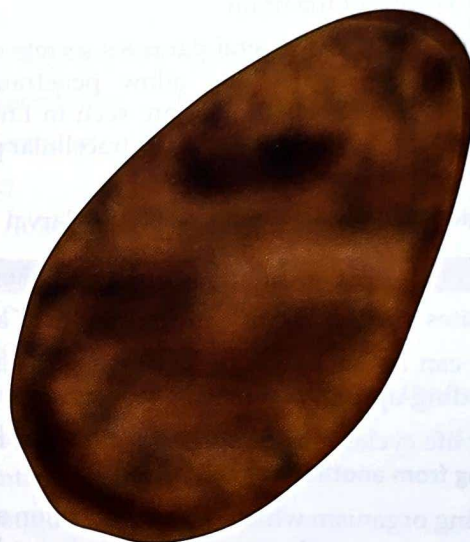


Figure 236: Lung flukes

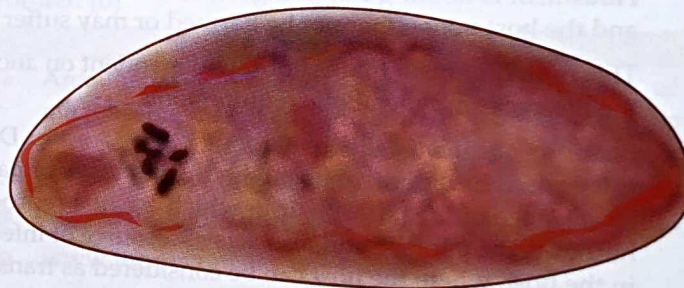


Figure 237: Intestinal flukes

Pathogenesis of parasitic diseases

- **Physical obstruction, irritation and trauma:** A high load of worms in the intestine can cause intestinal obstruction (*Ascaris*).
- Attachment to the intestinal mucosa causes irritation and bleeding from the villi (hookworm).
- Also, the migration of worms through the lungs as part of the natural life cycle causes mechanical irritation and damage to the pulmonary capillaries (e.g. *Strongyloides*).

Hookworm

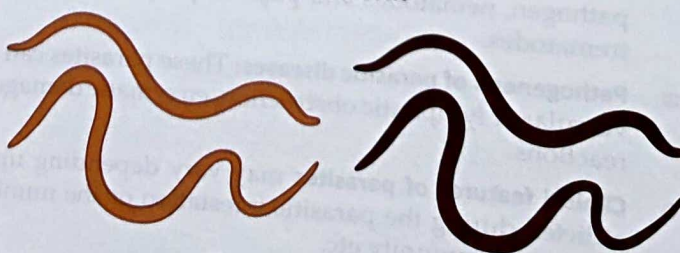


Figure 238: Hookworms

- **Vascular or lymphatic obstruction:** Elephantiasis –

oedematous swelling of the limbs – is caused by the blockage of the lymphatics by the adult worms in filariasis.

Cerebral malaria can occur due to the blockage of blood vessels of the brain.

- ⊗ **Enzymatic damage:** Several parasites secrete enzymes that damage tissues and allow penetration. For example, flask shaped ulcer are seen in *Entamoeba histolytica* infections of the gut. Intracellular parasites cause the necrosis of the cells.
- ⊗ **Allergic reactions:** The presence of the larval rms and

adult worm in the host elicits an increase in IgE production, which results in allergic manifestations. The rupture of hydatid cysts to release hydatid fluid can trigger an anaphylactic reaction.

- ⊗ **Inflammation:** Many parasites provoke cellular infiltration and proliferation, resulting in inflammatory reaction. Granuloma formation can be seen in some parasites such as *Entamoeba histolytica*.
- ⊗ **Neoplastic reactions:** Some parasites such as *Schistosoma* are associated with carcinoma.

Key summary of the chapter

- ⊗ Parasites are probably the most complex of all microbes infecting humankind. They can be **unicellular** organisms (e.g. *Plasmodium* species, which causes malaria) or **multicellular**, with sizes extending upto 10 meters (e.g. tapeworm). Their life cycles may be simple or complex, requiring multiple hosts and vectors. The literal meaning of parasite is 'eating from another's plate'.
- ⊗ A living organism which receives nutrition and shelter from another organism where it lives is called **parasite**. This need not necessarily cause disease in living beings. Parasitism is nothing but living in association with the host. The parasite derives all benefits from the association and the host may either not be harmed or may suffer the consequences of association. This is an organism that is entirely dependent on another organism, which is referred as its host for all or part of the life cycle and metabolic requirements.
- ⊗ **Host:** an organism which harbours the parasite. Depending on their relationship with parasite, host can be categorised as definitive host where adult stage of parasite lives or intermediate host where the host harbours the larval stage of the parasite or in which asexual reproduction takes place (protozoa), reservoir host where the host harbours the parasite and also acts as a source of infection or paratenic host where the parasite is unable to further in the host, where the host can be considered as transport host or carrier host.
- ⊗ **Habitat:** A place where the parasite naturally lives and grows is known as its habitat, e.g. malarial parasite in humans resides in the liver and RBC.
- ⊗ **Life cycle of parasites:** a number of parasites follow a simple or direct life cycle with a single host, some require two or more hosts to complete their life cycles.
- ⊗ **Classes of parasites:** unicellular (single celled parasite) and multicellular (having multiple cells)
- ⊗ **Classification of parasites** which can cause disease can be classified in three kingdoms –
 - (i) **Protozoa;**
 - (ii) **Stramenopila** and
 - (iii) **Animalia.**
 Helminths are categorised under animalia. These are further categorised into two phyla, namely, phylum nemathelminths which contain a single group of pathogen, nematodes and phylum platyhelminths which include two classes of pathogens, namely cystodes and trematodes.
- ⊗ **Pathogenesis of parasitic diseases:** These parasites can cause physical obstruction, irritation and trauma to the body, vascular or lymphatic obstruction, enzymatic damage to the tissues, allergic reactions, inflammation and neoplastic reactions.
- ⊗ **Clinical features of parasites** may vary depending upon the nature of parasite in the body, the part of the body afflicted during the parasitic infestation or the number of parasites afflicting the human body or the presence or absence of immunity etc.

Self-Assessment: Questions for Review

Short essay questions

- ⊛ Describe helminths
- ⊛ Describe parasites

MCQs

1. Plasmodium belongs to

- a. Protozoa; b. Nematodes;
- c. Trematodes; d. Cestodes

Answer: (a)

2. Which of the following is incorrect regarding protozoa?

- a. Unicellular;
- b. Multicellular;
- c. Eukaryotic cell;
- d. Microscopic parasite

Answer: (b)

3. Helminths belong to the group of

- a. Animalia; b. Birds;
- c. Reptiles; d. Amphibians

Answer: (a)

4. Hook worms afflict which part of the body?

- a. Large intestine;
- b. Lung parenchyma;
- c. Stomach;
- d. Urethra

Answer: (a)

5. Malaria is considered to be disease of

- a. Protozoa; b. Helminths;
- c. Nematodes; d. Cestodes

Answer: (a)

6. Tapeworms belong to the group of

- a. Nematodes; b. Cestodes;
- c. Trematodes; d. Protozoa

Answer: (b)

7. Endo parasite lives in the following except

- a. Lymph; b. Blood;
- c. Muscle; d. Skin

Answer: (d)

8. Taenia saginata belongs to the group of

- a. Nematodes; b. Trematodes;
- c. Cestodes; d. Protozoa

Answer: (c)

9. Liver flukes belong to the group of

- a. Nematodes; b. Trematodes;
- c. Cestodes; d. Protozoa

Answer: (b)

10. Sleeping sickness is caused by which of the following?

- a. Plasmodium species;
- b. Entamoeba histolytica;
- c. Round worm;
- d. Hook worm

Answer: (b)

Chapter resources

- ⊛ Ananthanarayan and Paniker's Textbook of microbiology, 12th edition, edited by Reba Kanungo and Sonal Saxena, Published by Universities Press (India) Private Limited, Hyderabad, 2022
- ⊛ www.wikipedia.org
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